

Batch script: slurm_pbmc10k_coarse.sh

```
#!/bin/bash -l
#SBATCH -J BMI500-10k-coarse
#SBATCH -p batch
#SBATCH -t 1-0:0:0
#SBATCH -o /opt/scratchspace/jangwon/bmi500/log/pbmc10k-coarse-%A.out
#SBATCH -e /opt/scratchspace/jangwon/bmi500/log/pbmc10k-coarse-%A.err
#SBATCH --mem=32G
#SBATCH -n 1
#SBATCH -c 12
#SBATCH -N 1
#SBATCH --mail-type=ALL --mail-user=jlim249@emory.edu

set -euo pipefail
cd /opt/scratchspace/jangwon/bmi500
MINICONDA_PATH=/home/jlim249/miniconda3
export PATH="/home/jlim249/miniconda3/bin:$PATH"
source "$MINICONDA_PATH/etc/profile.d/conda.sh"
conda activate bmi500
export PYTHONNOUSERSITE=1
mkdir -p ./output/coarse
/usr/bin/time -v /home/jlim249/miniconda3/envs/bmi500/bin/python scanpy_pbmc.py --data-set pbmc10k
--out-dir ./output/coarse --num-threads 12
```

Batch script: slurm_pbmc10k_fine.sh

```
#!/bin/bash -l
#SBATCH -J BMI500-10k-fine
#SBATCH -p batch
#SBATCH -t 1-0:0:0
#SBATCH -o /opt/scratchspace/jangwon/bmi500/log/pbmc10k-fine-%A.out
#SBATCH -e /opt/scratchspace/jangwon/bmi500/log/pbmc10k-fine-%A.err
#SBATCH --mem=32G
#SBATCH -n 1
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#SBATCH -N 1
#SBATCH --mail-type=ALL --mail-user=jlim249@emory.edu

set -euo pipefail
cd /opt/scratchspace/jangwon/bmi500
MINICONDA_PATH=/home/jlim249/miniconda3
export PATH="/MINICONDA_PATH/bin:$PATH"
source "$MINICONDA_PATH/etc/profile.d/conda.sh"
conda activate bmi500
export PYTHONNOUSERSITE=1
mkdir -p ./output/fine
/home/jlim249/miniconda3/envs/bmi500/bin/python scanpy_pbmc.py --data-set pbmc10k --out-dir ./output/fine
--num-threads 12 --profile-output ./output/fine/pbmc10k-rank_genes_groups.prof
```

Batch script: slurm_pbmc10k_instrumented.sh

```
#!/bin/bash -l
#SBATCH -J BMI500-10k-instr
#SBATCH -p batch
#SBATCH -t 1-0:0:0
#SBATCH -o /opt/scratchspace/jangwon/bmi500/log/pbmc10k-instr-%A.out
#SBATCH -e /opt/scratchspace/jangwon/bmi500/log/pbmc10k-instr-%A.err
#SBATCH --mem=32G
#SBATCH -n 1
#SBATCH -c 12
#SBATCH -N 1
#SBATCH --mail-type=ALL --mail-user=jlim249@emory.edu

set -euo pipefail
cd /opt/scratchspace/jangwon/bmi500
MINICONDA_PATH=/home/jlim249/miniconda3
export PATH="$MINICONDA_PATH/bin:$PATH"
source "$MINICONDA_PATH/etc/profile.d/conda.sh"
conda activate bmi500
export PYTHONNOUSERSITE=1
mkdir -p ./output/instrumented
/home/jlim249/miniconda3/envs/bmi500/bin/python scanpy_pbmc.py --data-set pbmc10k --out-dir
./output/instrumented --num-threads 12
```

Batch script: slurm_pbmc3k_coarse.sh

```
#!/bin/bash -l
#SBATCH -J BMI500-3k-coarse
#SBATCH -p batch
#SBATCH -t 1-0:0:0
#SBATCH -o /opt/scratchspace/jangwon/bmi500/log/pbmc3k-coarse-%A.out
#SBATCH -e /opt/scratchspace/jangwon/bmi500/log/pbmc3k-coarse-%A.err
#SBATCH --mem=16G
#SBATCH -n 1
#SBATCH -c 4
#SBATCH -N 1
#SBATCH --mail-type=ALL --mail-user=jlim249@emory.edu

set -euo pipefail
cd /opt/scratchspace/jangwon/bmi500
MINICONDA_PATH=/home/jlim249/miniconda3
export PATH="/opt/scratchspace/jangwon/bmi500/miniconda3/bin:$PATH"
source "$MINICONDA_PATH/etc/profile.d/conda.sh"
conda activate bmi500
export PYTHONNOUSERSITE=1
mkdir -p ./output/coarse
/usr/bin/time -v /home/jlim249/miniconda3/envs/bmi500/bin/python scanpy_pbmc.py --data-set pbmc3k
--out-dir ./output/coarse --num-threads 4
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source "$MINICONDA_PATH/etc/profile.d/conda.sh"
conda activate bmi500
export PYTHONNOUSERSITE=1
mkdir -p ./output/fine
/home/jlim249/miniconda3/envs/bmi500/bin/python scanpy_pbmc.py --data-set pbmc3k --out-dir ./output/fine
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Batch script: slurm_pbmc3k_instrumented.sh

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/home/jlim249/miniconda3/envs/bmi500/bin/python scanpy_pbmc.py --data-set pbmc3k --out-dir
./output/instrumented --num-threads 4
```

Batch script: slurm_pbmc6k_coarse.sh

```
#!/bin/bash -l
#SBATCH -J BMI500-6k-coarse
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#SBATCH -t 1-0:0:0
#SBATCH -o /opt/scratchspace/jangwon/bmi500/log/pbmc6k-coarse-%A.out
#SBATCH -e /opt/scratchspace/jangwon/bmi500/log/pbmc6k-coarse-%A.err
#SBATCH --mem=24G
#SBATCH -n 1
#SBATCH -c 8
#SBATCH -N 1
#SBATCH --mail-type=ALL --mail-user=jlim249@emory.edu

set -euo pipefail
cd /opt/scratchspace/jangwon/bmi500
MINICONDA_PATH=/home/jlim249/miniconda3
export PATH="/opt/scratchspace/jangwon/bmi500/miniconda3/bin:$PATH"
source "$MINICONDA_PATH/etc/profile.d/conda.sh"
conda activate bmi500
export PYTHONNOUSERSITE=1
mkdir -p ./output/coarse
/usr/bin/time -v /home/jlim249/miniconda3/envs/bmi500/bin/python scanpy_pbmc.py --data-set pbmc6k
--out-dir ./output/coarse --num-threads 8
```

Batch script: slurm_pbmc6k_fine.sh

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source "$MINICONDA_PATH/etc/profile.d/conda.sh"
conda activate bmi500
export PYTHONNOUSERSITE=1
mkdir -p ./output/fine
/home/jlim249/miniconda3/envs/bmi500/bin/python scanpy_pbmc.py --data-set pbmc6k --out-dir ./output/fine
--num-threads 8 --profile-output ./output/fine/pbmc6k-rank_genes_groups.prof
```


Batch script: slurm_pbmc6k_instrumented.sh

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#!/bin/bash -l
#SBATCH -J BMI500-6k-instr
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cd /opt/scratchspace/jangwon/bmi500
MINICONDA_PATH=/home/jlim249/miniconda3
export PATH="/MINICONDA_PATH/bin:PATH"
source "$MINICONDA_PATH/etc/profile.d/conda.sh"
conda activate bmi500
export PYTHONNOUSERSITE=1
mkdir -p ./output/instrumented
/home/jlim249/miniconda3/envs/bmi500/bin/python scanpy_pbmc.py --data-set pbmc6k --out-dir
./output/instrumented --num-threads 8
```

Key Performance Findings

KEY PERFORMANCE FINDINGS

Coarse-grain usage (wall / user CPU / system CPU / maximum memory):

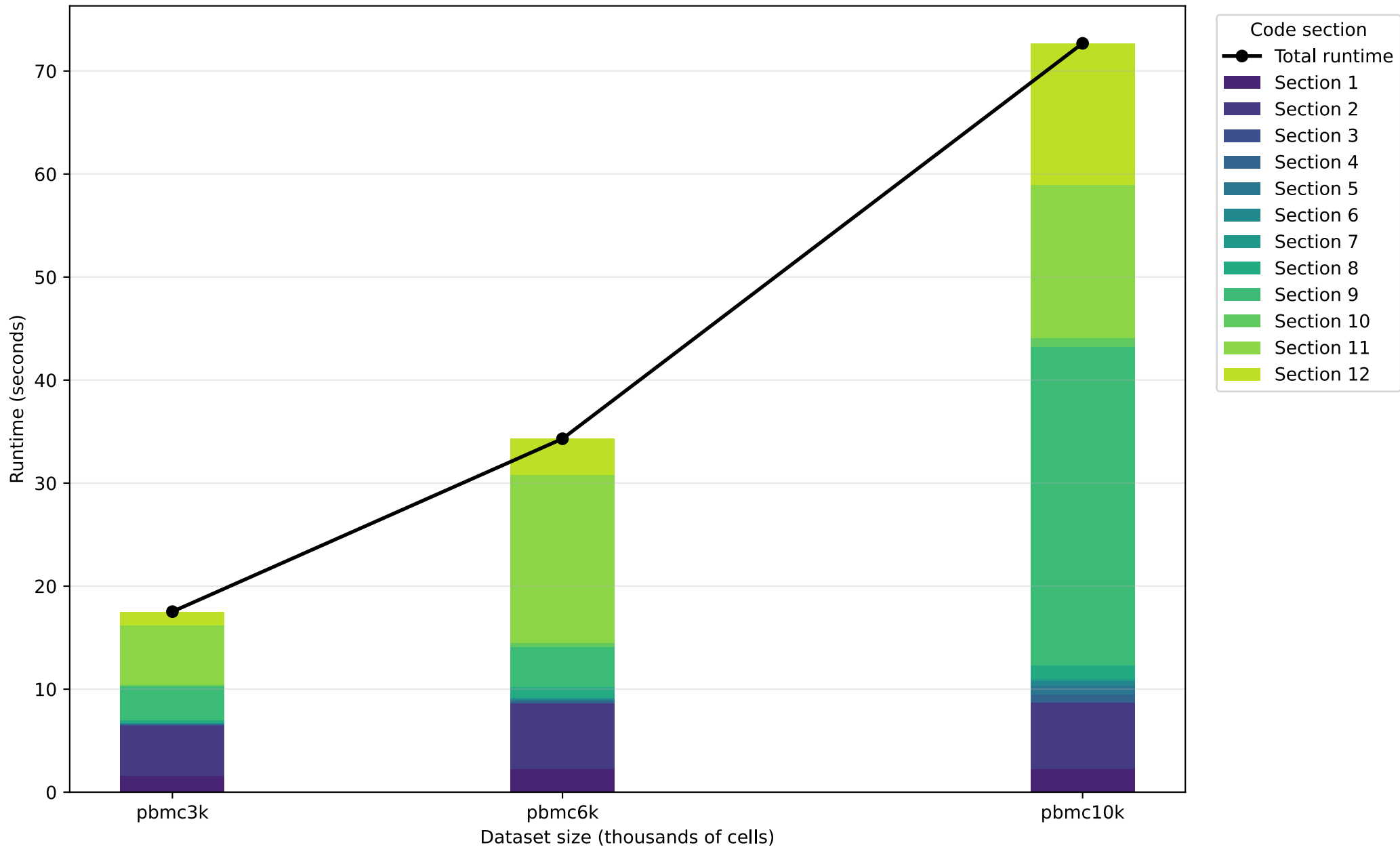
pbmc3k: 0:19.52 / 18.76 s / 1.60 s / 868 MB

pbmc6k: 0:26.62 / 34.84 s / 2.49 s / 962 MB

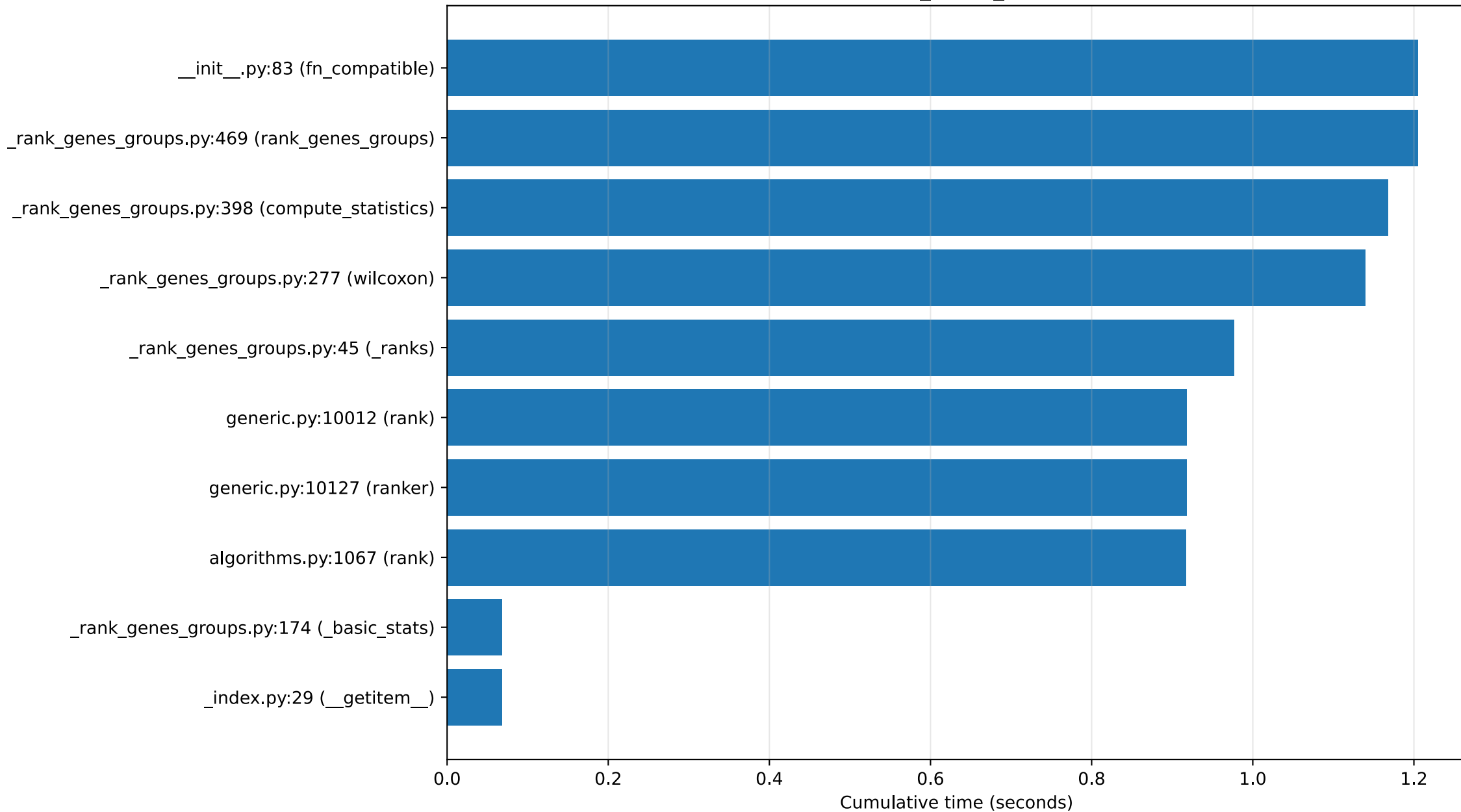
pbmc10k: 0:56.27 / 63.20 s / 6.63 s / 1846 MB

I believe the major bottleneck happens at the section 9 as shown in the stacked bar chart. Runtime grew significantly from the smaller datasets, so if the data size were to grow to 20K, the expected running times would be approximately 2.5 times. Since the entire runtime for 10K dataset is around 60-70 seconds, the expected runtime would be around 150-175 seconds. If it's not linear, which is most likely, then it might take much longer.

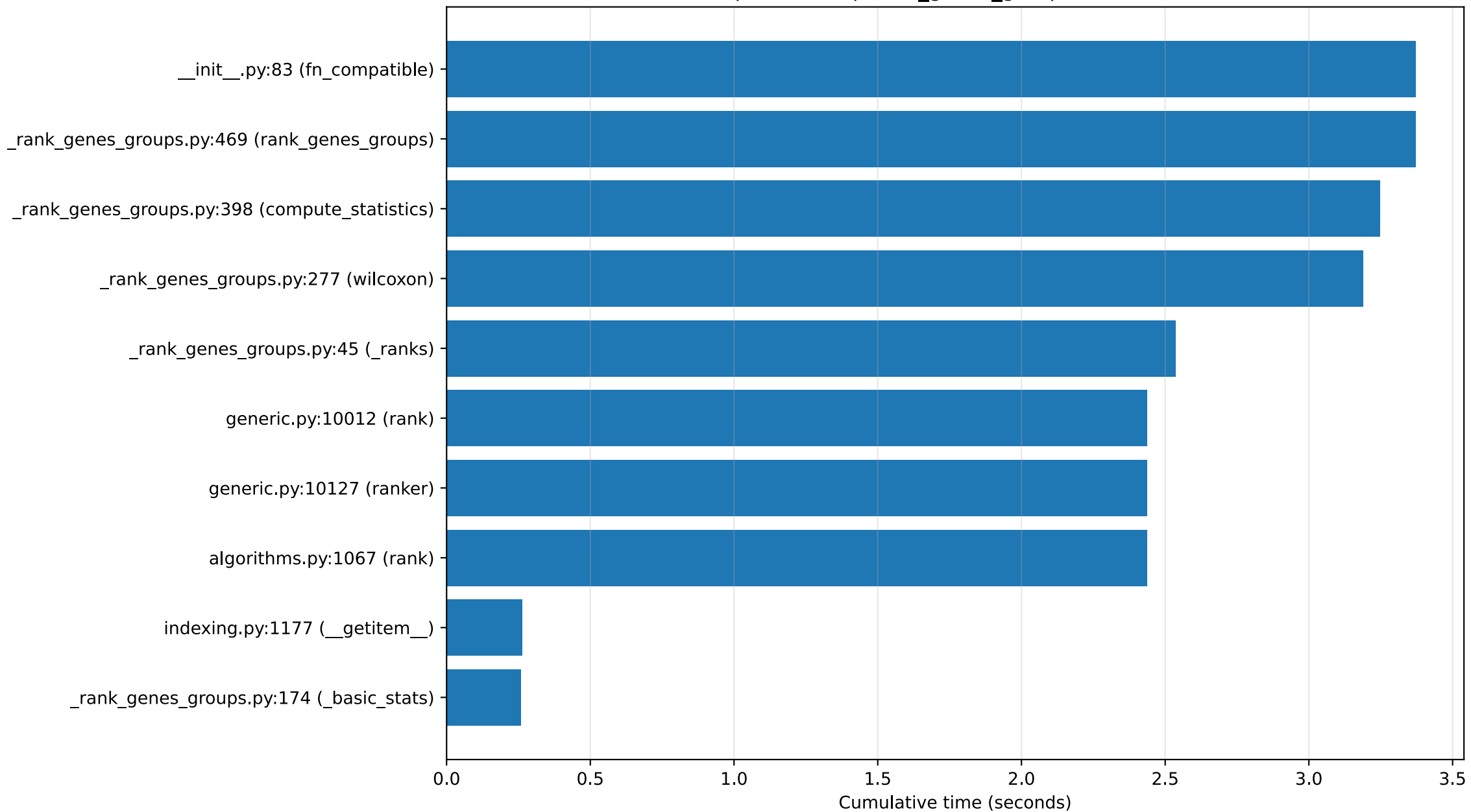
Instrumented runtime growth by dataset



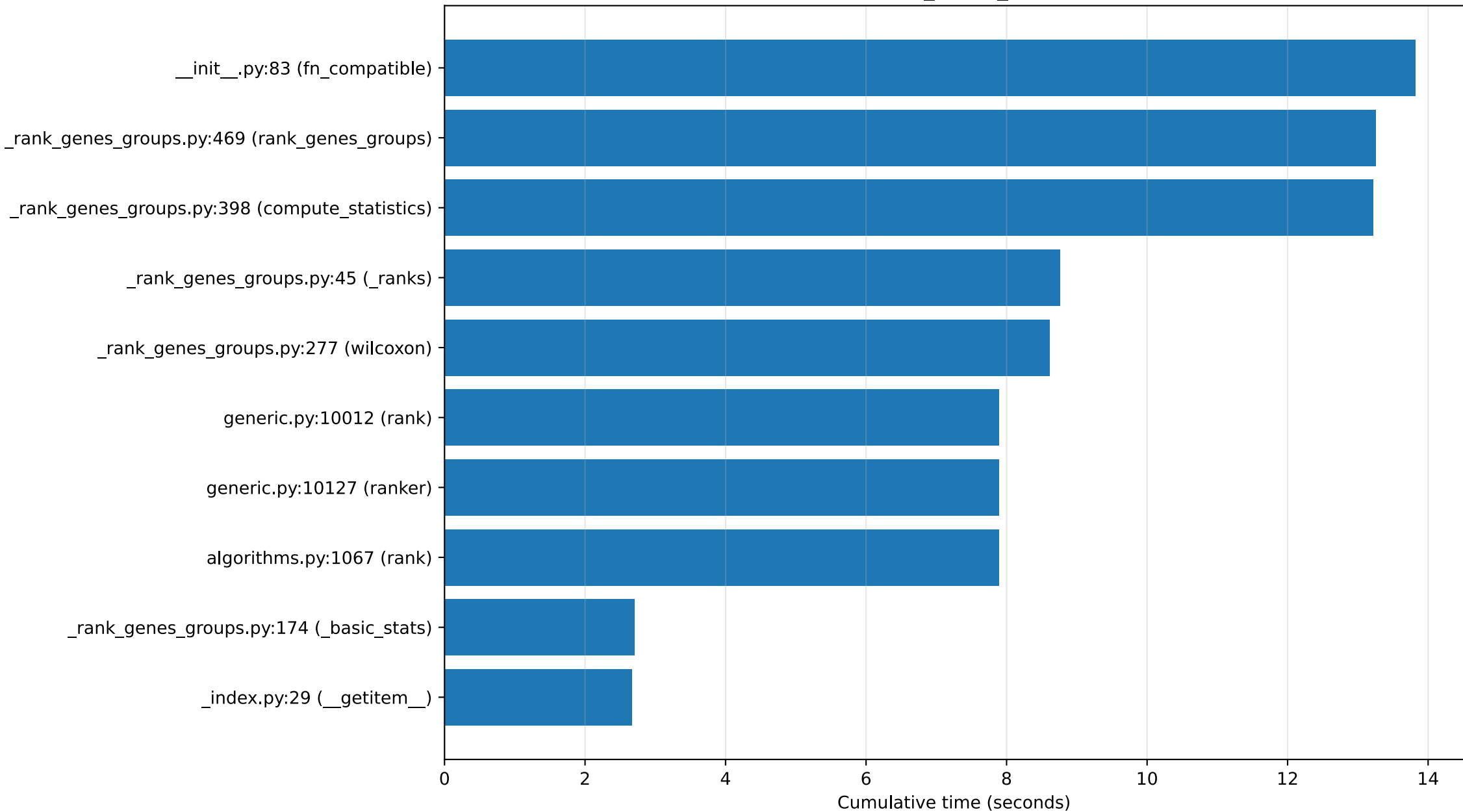
pbmc3k: top rank_genes_groups functions



pbmc6k: top_rank_genes_groups functions



pbmc10k: top rank_genes_groups functions



pbmc3k: coarse profiling and instrumented timings

Dataset: pbmc3k

Coarse profiling output after Command being timed:

```
"/home/jlim249/miniconda3/envs/bmi500/bin/python scanpy_pbmc.py --data-set pbmc3k --out-dir
./output/coarse --num-threads 4"
User time (seconds): 18.76
System time (seconds): 1.60
Percent of CPU this job got: 104%
Elapsed (wall clock) time (h:mm:ss or m:ss): 0:19.52
Average shared text size (kbytes): 0
Average unshared data size (kbytes): 0
Average stack size (kbytes): 0
Average total size (kbytes): 0
Maximum resident set size (kbytes): 888488
Average resident set size (kbytes): 0
Major (requiring I/O) page faults: 2775
Minor (reclaiming a frame) page faults: 153289
Voluntary context switches: 15803
Involuntary context switches: 372
Swaps: 0
File system inputs: 588064
File system outputs: 86512
Socket messages sent: 0
Socket messages received: 0
Signals delivered: 0
Page size (bytes): 4096
Exit status: 0
```

Instrumented timings:

```
section 1 elapsed: 1.590 seconds
section 2 elapsed: 4.912 seconds
section 3 elapsed: 0.001 seconds
section 4 elapsed: 0.102 seconds
section 5 elapsed: 0.047 seconds
section 6 elapsed: 0.020 seconds
section 7 elapsed: 0.030 seconds
section 8 elapsed: 0.252 seconds
section 9 elapsed: 3.348 seconds
section 10 elapsed: 0.088 seconds
section 11 elapsed: 5.828 seconds
section 12 elapsed: 1.307 seconds
```

pbmc6k: coarse profiling and instrumented timings

Dataset: pbmc6k

Coarse profiling output after Command being timed:

```
"/home/jlim249/miniconda3/envs/bmi500/bin/python scanpy_pbmc.py --data-set pbmc6k --out-dir
./output/coarse --num-threads 8"
User time (seconds): 34.84
System time (seconds): 2.49
Percent of CPU this job got: 140%
Elapsed (wall clock) time (h:mm:ss or m:ss): 0:26.62
Average shared text size (kbytes): 0
Average unshared data size (kbytes): 0
Average stack size (kbytes): 0
Average total size (kbytes): 0
Maximum resident set size (kbytes): 984928
Average resident set size (kbytes): 0
Major (requiring I/O) page faults: 2805
Minor (reclaiming a frame) page faults: 215875
Voluntary context switches: 19308
Involuntary context switches: 237243
Swaps: 0
File system inputs: 672344
File system outputs: 197560
Socket messages sent: 0
Socket messages received: 0
Signals delivered: 0
Page size (bytes): 4096
Exit status: 0
```

Instrumented timings:

```
section 1 elapsed: 2.253 seconds
section 2 elapsed: 6.402 seconds
section 3 elapsed: 0.001 seconds
section 4 elapsed: 0.184 seconds
section 5 elapsed: 0.153 seconds
section 6 elapsed: 0.090 seconds
section 7 elapsed: 0.079 seconds
section 8 elapsed: 1.073 seconds
section 9 elapsed: 3.878 seconds
section 10 elapsed: 0.378 seconds
section 11 elapsed: 16.326 seconds
section 12 elapsed: 3.491 seconds
```


pbmc10k: coarse profiling and instrumented timings

Dataset: pbmc10k

Coarse profiling output after Command being timed:

```
"/home/jlim249/miniconda3/envs/bmi500/bin/python scanpy_pbmc.py --data-set pbmc10k --out-dir
./output/coarse --num-threads 12"
User time (seconds): 63.20
System time (seconds): 6.63
Percent of CPU this job got: 124%
Elapsed (wall clock) time (h:mm:ss or m:ss): 0:56.27
Average shared text size (kbytes): 0
Average unshared data size (kbytes): 0
Average stack size (kbytes): 0
Average total size (kbytes): 0
Maximum resident set size (kbytes): 1890192
Average resident set size (kbytes): 0
Major (requiring I/O) page faults: 40
Minor (reclaiming a frame) page faults: 953916
Voluntary context switches: 24196
Involuntary context switches: 409586
Swaps: 0
File system inputs: 628960
File system outputs: 831456
Socket messages sent: 0
Socket messages received: 0
Signals delivered: 0
Page size (bytes): 4096
Exit status: 0
```

Instrumented timings:

```
section 1 elapsed: 2.275 seconds
section 2 elapsed: 6.459 seconds
section 3 elapsed: 0.001 seconds
section 4 elapsed: 0.757 seconds
section 5 elapsed: 0.873 seconds
section 6 elapsed: 0.448 seconds
section 7 elapsed: 0.176 seconds
section 8 elapsed: 1.308 seconds
section 9 elapsed: 30.960 seconds
section 10 elapsed: 0.803 seconds
section 11 elapsed: 14.895 seconds
section 12 elapsed: 13.733 seconds
```